

NEUTRINO OSCILLATION EXPERIMENT CONTROL

WHAT IS A NEUTRINO?

Neutrinos are small particles that only interact with other particles through the weak force.

There are three flavours of neutrino: electron, muon and tau. Each type of neutrino also has a corresponding anti-neutrino, which might act differently.

ν_τ

ν_e

ν_μ

ν_1

WHY DO THEY OSCILLATE?

Each flavour of neutrino is a specific mixture of the three different neutrino mass states. As they travel, the mixture of mass states changes slightly, so when the neutrino is finally detected, there is a chance that it will have changed flavour.

ν_2

ν_3

HOW ARE THEY MEASURED?

The Deep Underground Neutrino Experiment will send neutrinos 1300km through the earth from Fermilab to Sanford Underground Research Facility. By checking how many neutrinos of different flavours and energies arrive at Sanford, particle physicists can learn more about the mass states, the difference between neutrinos and anti-neutrinos, and how the much of each mass is in each flavour mixture.

WHY DOES THIS MATTER?

Learning more about neutrino oscillations will help to answer fundamental physics questions such as:

WHY IS THE
UNIVERSE MADE OF
MATTER AND NOT
ANTIMATTER?

HOW DO BLACK
HOLES AND
SUPERNOVAE
OCCUR?

WHAT IS
BEYOND THE
STANDARD MODEL
OF PHYSICS?

HOW TO PLAY

1. CHOOSE YOUR PARAMETERS

DM32

Change the difference
between mass state 2 and 3

TH23

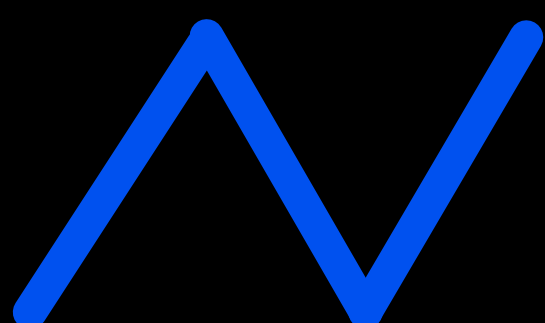
Change the mixture of mass
states in electron neutrinos

DCP

Make anti-neutrinos behave
differently to neutrinos

BONUS: +1X FOR EVERY PARAMETER

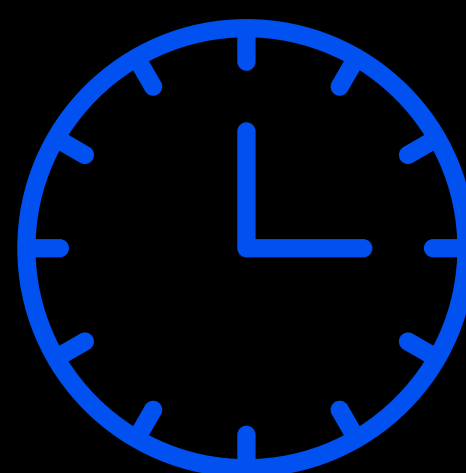
2. CHOOSE YOUR MODES



Noise

Detectors aren't perfect —
sometimes they miss
neutrinos or make them up.

BONUS: 1.5X SCORE

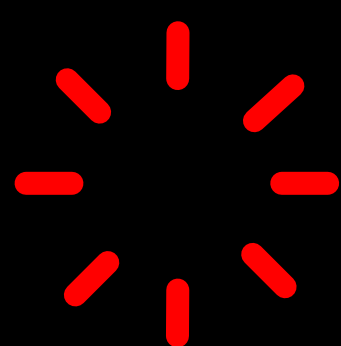


Real Time Data

Neutrino events are really rare — try
to match your model to the data while
the new events occur

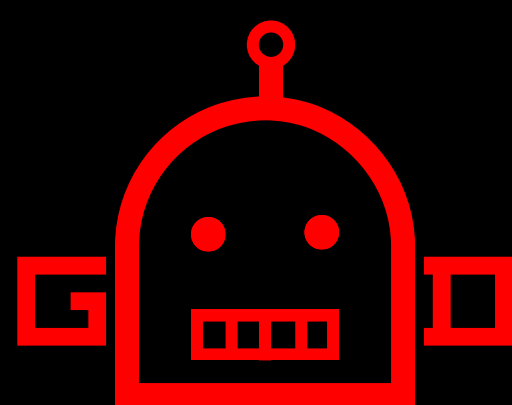
BONUS: 2X SCORE

3. CHOOSE YOUR AI OPPONENT



NONE

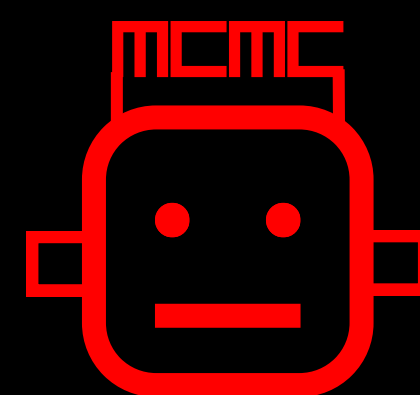
Just you vs the clock. No
distractions.



GRADIENT DESCENT

Constantly jumps in the direction
of a better answer, trying to find
the lowest likelihood spot it can
reach

BONUS: 2X SCORE

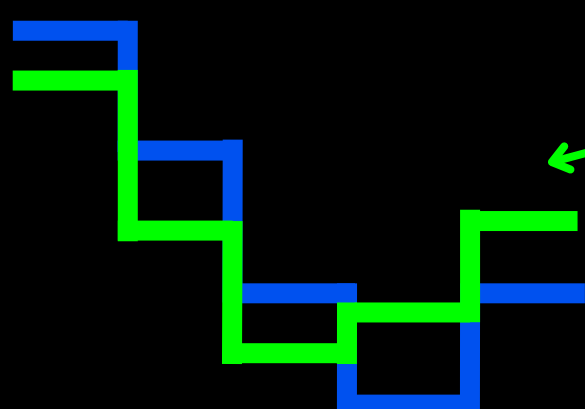


MARKOV CHAIN MONTE
CARLO

Stalks around the space, picking new
spots based on a combination of their
likelihood and random chance.

BONUS: 2X SCORE

4. MATCH THE DATA



← Your model

← Detector Data

Change the parameters until your model matches the data from the
detector — then hit the big red button to finalise your model and get
your score!

To get the highest score, get as close to the data as you can, as fast
as you can!

